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Multistage Stochastic Optimization for Production-Inventory Planning with Intermittent Renewable Energy

Mehdi Golari, Neng Fan
Department of Systems and Industrial Engineering, University of Arizona, Tucson, Arizona 85721, USA, golari@email.arizona.edu, nfan@email.arizona.edu

Tongdan Jin
Ingram School of Engineering, Texas State University, San Marcos, Texas 78666, USA, tj17@txstate.edu

A growing number of companies install wind and solar generators in their energy-intensive facilities to attain low-carbon manufacturing operations. However, there is a lack of methodological studies on operating large manufacturing facilities with intermittent power. This study presents a multi-period, production-inventory planning model in a multi-plant manufacturing system powered with onsite and grid renewable energy. Our goal is to determine the production quantity, the stock level, and the renewable energy supply in each period such that the aggregate production cost (including energy) is minimized. We tackle this complex decision problem in three steps. First, we present a deterministic planning model to attain the desired green energy penetration level. Next, the deterministic model is extended to a multistage stochastic optimization model taking into account the uncertainties of renewables. Finally, we develop an efficient modified Benders decomposition algorithm to search for the optimal production schedule using a scenario tree. Numerical experiments are carried out to verify and validate the model integrity, and the potential of realizing high-level renewables penetration in large manufacturing system is discussed and justified.

Key words: green energy coefficient; onsite renewable generation; production planning; Benders decomposition
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1. Introduction

Recently a growing number of companies are tapping into onsite renewable energy sources to lower carbon footprint of their energy-intensive facilities. For instance, Honda installed two wind turbines (WT) at its auto transmission plant in Russells Point, Ohio. Both turbines can contribute up to 10% of the plant's total electricity usage (Sinpetru 2014). Anheuser-Busch installed a 1.5 MW WT and 1.2 MW solar photovoltaics (PV) arrays nearby the factory in Fairfield, CA, providing 24% of green electricity to the brewery facility (Eberling 2013). Apple built the nation's largest onsite generation system, consisting of 40 MW PV arrays and 10 MW biogas fuel cell system, to power its data center in Maiden, NC (Apple 2012). These firms along with others see integration of onsite renewable energy as a competitive edge by offering green quality products to their customers.

In addition to the environmental benefits, integrating renewable energy in industrial and public facilities could be seen as part of the smart grid initiatives, accelerating the adoption of distributed generation (DG) systems. Unlike centralized generation, a DG system produces electricity from many small and dispersed energy resources (DER) that are installed in close proximity to the end users. Typical DER units include WT, solar PV, fuel cells, micro-turbines, and diesel engines. DG not only reduces the carbon emissions and diversifies the energy portfolio, but also relieves the transmission bottlenecks during peak hours. To facilitate the penetration of renewable DG systems, several roadblocks, such as unpredictability and intermittency of the wind and solar generation, must be removed (Wee et al. 2012).

This study aims to design a low-carbon production-inventory system by integrating onsite and grid renewable energy into a multi-plant manufacturing environment. We optimize the production and inventory over a multi-period horizon such that the total cost (including energy) is minimized while the green energy coefficient (GEC) target is ensured. GEC is defined as the ratio of the renewable energy over the total energy consumed, and is becoming a hallmark to assess the manufacturing sustainability. We tackle this complex production-inventory planning problem in

three steps. First, a deterministic production model incorporating wind and solar energy is formulated and solved to meet the GEC target. Second, the deterministic model is extended to a stochastic optimization model that captures the intermittent nature of renewable power. Finally, we present an efficient modified Benders decomposition algorithm to solve the multiperiod stochastic production planning model.

The contribution of this study lies in three aspects. First, we take an early step to incorporate both onsite and grid wind and solar energy into the production-inventory model to achieve low-carbon, cost-effective manufacturing operation. Our model broadens the production planning literature by leveraging renewables as an indispensable source to attain manufacturing sustainability target. Second, we propose to use GEC as a green quality criterion to gauge the eco-responsibility of a product. We are not aware if this basic environmental performance index has ever been considered in existing production and inventory literature. GEC motivates an environmentally cautious manufacturer to trump his competitors by offering a product with a smaller or even zero carbon footprint. Third, we develop an efficient modified Benders decomposition algorithm to search for the optimal production schedule using a scenario tree that takes into account renewable generation uncertainty.

The remainder of the study is presented as follows. Section 2 reviews the literature about onsite generation, production-inventory planning, and stochastic optimization techniques. Section 3 presents a multiplant deterministic production model with wind and solar energy integration. Section 4 extends the deterministic model to multistage stochastic programming model considering variable energy supply. Solution algorithms based on modified Benders decomposition are developed in section 5. In section 6, numerical experiments are provided to demonstrate the performance of the proposed model. Section 7 concludes the study.

2. Literature Review

In this section, we review the state-of-the-art pertaining to stochastic production-inventory planning, design for low-carbon manufacturing systems, and stochastic optimization algorithms.

2.1. Stochastic Production-Inventory Model

Stochastic production-inventory models with various internal and external uncertainties have been extensively studied in literature. For a comprehensive review on this research stream, readers are referred to Yano and Lee (1995) and Mula et al. (2006). Below we revisit several models that are closely related to our work.

Federgruen and Zipkin (1984) optimize a stochastic production program for a central plant that fulfills random demands from several locations. They minimize the expected holding and backorder cost over a finite number of time periods. Lee and Yano (1988) develop a back-track dynamic programming method to find the optimality of a serial production system with the goal to minimize operating, holding, and backorder costs subject to random yield. Wang and Gerchak (1996) further generalize the random yield model by incorporating capacity variability. Kira et al. (1997) propose a hierarchical method to optimize a multi-period, multi-product production scheduling problem under a finite set of demands using stochastic linear programming. Armentano et al. (2001) address the lot-sizing problem in capacitated multistage systems by minimizing production, inventory, and setup costs based on the Wagner–Whitin algorithm. Duran et al. (2007) design an optimal production and inventory policy assuming general stochasticity for demand and production. Kogan (2009) investigates a basic production planning problem under multiplicative stochastic yield and additive stochastic demand. Singer and Khmelnitsky (2010) consider optimal production control of a one-product-type production inventory system where demand is a discrete-time stochastic process.

Recently, Higle and Kempf (2011) propose a stochastic programming model for production planning under both supply and demand uncertainty via Markov decision process. In Geunes et al. (2011), approximation algorithms for more generalized class of supply chain and logistic models under deterministic and stochastic demands are studied. Cheng et al. (2012) optimize a hybrid two-stage push–pull production system with uncertain peak demand subject to the service level constraints. Zhang et al. (2012) investigate a multi-echelon uncapacitated lot-sizing problem where the intermediate echelons face uncertain internal and external demands. With the growing penetration of wind and solar energy in the utility market, it is also imperative to incorporate the energy supply variability and the carbon constraints into the production-inventory management program to lower the manufacturing cost with superior environmental performance.

2.2. Energy-Aware and Low-Carbon Manufacturing

Some attempts have been made to design the production program considering the electricity price elasticity. Ierapetritou et al. (2002) optimize the production rate, plant operation modes, and inventory level on an hourly basis to meet the demand in the air separation industry. They construct a two-stage stochastic programming model to minimize the plant's

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operating costs subject to real-time electricity rate. Karwan and Keblis (2007) solve a similar planning problem using simulation-based optimization via a rolling horizon method. Mitra et al. (2012) develop a deterministic formulation that can efficiently handle the transient plant behavior by taking into account the variations of seasonal electricity tariffs (i.e., time-of-use). Our study resembles theirs in the sense that we also seek an optimal production-inventory policy with uncertain energy availability. A main difference is that we propose to integrate wind and solar energy into large manufacturing enterprises to achieve eco-friendly operation.

To mitigate environmental impacts, some researchers advocate direct integration of onsite and grid renewable energy into the manufacturing facility. Villarreal et al. (2013) investigate the integration of onsite solar and wind power into large wafer production systems. They analyze the loss-of-load probability and calculate the payback period for DG investment, and conclude that it is cost-effective if the average wind speed is above 5 m/s or the yearly overcast days are less than 35%. More recently, Li et al. (2016) propose a grid-connected DG system to realize net-zero carbon manufacturing operation using onsite renewable power. The study shows that dynamic pricing such as time-of-use and power-of-use has little impact on the levelized cost of onsite wind generation. In both studies, the variation of production and inventory on power demand is not explicitly considered.

Another research stream designs multi-period, production-inventory systems considering emissions trading and cap scheme. Letmathe and Balakrishnan (2005) integrate emission concerns into a production planning model under environmental legislations. Technologies that differ in their emission and resources consumption rate are factored into the objective function to maximize the total manufacturing profit. Chen and Monahan (2010) optimize the production and inventory under stringent pollution control with demand and environmental uncertainties. An environmental safety stock is built for the periods when the production rate cannot satisfy demand due to emission constraints. More recently, Zhang et al. (2016) assess the trade-off between carbon footprint and different production technologies, and propose an optimal production-inventory policy to lower the total cost in the presence of uncertain carbon allowance and trading market. Emissions quota restrictions have also been considered in power systems modeling under uncertain renewable energy. For example, Rebennack et al. (2012) present a multistage stochastic programming model to incorporate greenhouse gas emissions quotas for a hydro-thermal generation scheduling problem. In Rebennack (2014), a multistage stochastic model to seek optimal generation expansion decisions under emissions quota constraints is proposed.

2.3. Stochastic Programming Models and Algorithms

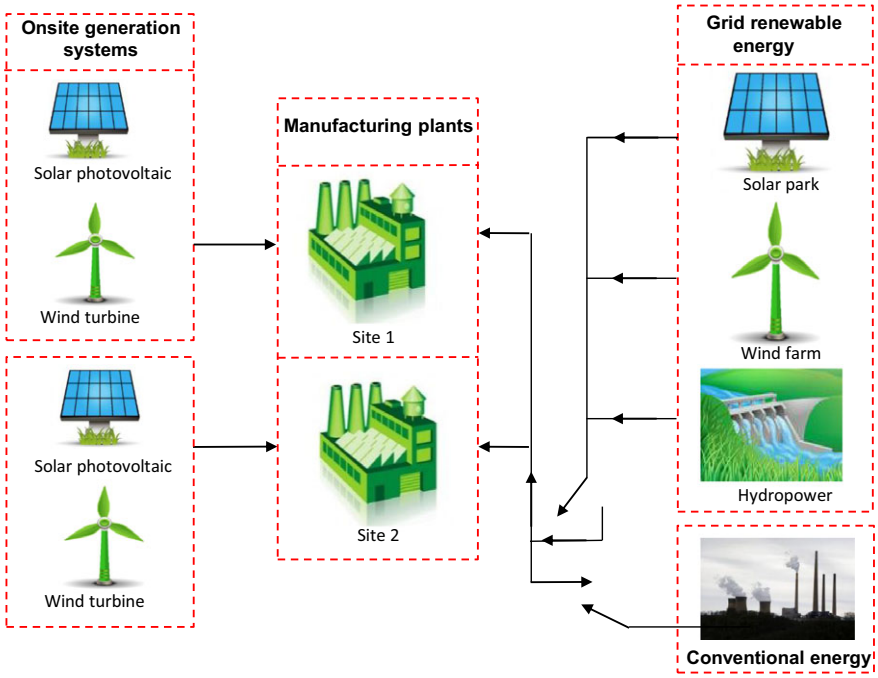
Multistage stochastic programming models have been extensively studied and applied to solve production-inventory models, generation expansion and other large-scale decision-makings (see Birge and Louveaux 2011, Kall and Mayer 2011). A multistage model allows several layers of decisions, where at each stage more random outcomes are realized. The uncertainties over a planning horizon with multiple periods can be captured by a scenario tree (see Birge and Louveaux 2011, Kall and Mayer 2011). If the objective function and all constraints are linear with no integer variables, the multistage decision model becomes a multistage stochastic linear program (MSLP). Although MSLP can be reformulated as an array of linear programs, the number of constraints grows exponentially with the number of stages. This makes this type of problem computationally intractable even if the number of random variables is finite. Therefore, in addition to deterministic solution methods such as Nested Benders Decomposition of Birge (1982) and Gassmann (1990), several approximation methods are proposed to solve an MSLP model if the problem size and the number of stages becomes large (see Sen and Zhou 2014). Recently, Rebennack (2016) embeds the stochastic dual dynamic programming into the scenario tree framework, and develops nested Benders decomposition algorithms successfully incorporating both scenario trees and sampling approaches for stage-wise independent and dependent processes.

3. Deterministic Production Planning with Renewables Integration

The goal of production planning is to determine the manufacturing schedule such that the demand over a time horizon is satisfied with minimum cost. In the global marketplace, inventory pooling and aggregate production are widely used to cope with demand uncertainty and mass customization. As a special resource for manufacturing activity, electricity has long been assumed to be available at any random point in time. With the growing penetration of wind and solar power, both the reliability and the security of renewable energy become a serious concern to utility companies. Thus, wind and solar energy availability should be treated as an uncertain parameter over the planning horizon, and its impact on the production-inventory decision needs to be explicitly

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Figure 1 A Multi-Plant Manufacturing System with Renewable Energy Integration



considered for the sake of operational stability. We classify the electric power into three categories based on the generation sources: (1) onsite renewable energy, (2) grid renewable energy, and (3) grid conventional energy. Energy in category 1 is produced by onsite WT and PV units at the manufacturing plant. Energy in category 2 is generated by hydro generators, large wind farms or solar parks. Energy in category 3 is generated by fossil fuel fired power plants. Figure 1 depicts a multi-plant manufacturing system powered by mixed energy sources consisting of wind, solar, hydro and conventional generators.

To facilitate the problem formulation, parameters and variables pertaining to the multi-plant manufacturing system are provided in Tables 1–3. In particular, Table 1 lists the sets and indices for manufacturing plants, product types, time periods, materials and labor, and energy sources. Table 2 summarizes the costs associated with production, inventory holding, backorders, energy as well as the product demand and capacity limits. Table 3 lists the decision variables, including the production quantity, the inventory level, and the backorders at different plants. The amounts of energy per category consumed at different plants are also listed in Table 3.

Below, we present a multi-plant manufacturing system where each plant is penetrated by intermittent onsite and grid renewable energy, and all plants are geographically dispersed. In our formulation, each plant is capable of producing multiple product types

Table 1 Sets and Indices

Symbol	Description
M	Set of manufacturing plants
I	Set of product types
T	Set of time periods
K	Set of resources for materials and labor
R	Set of renewable energy resourses
m	Index for manufacturing plants, and $m = 1, 2, \ldots, \|M\|$
i	Index for product type, and $i = 1, 2, \ldots, \|I\|$
t	Index for period, and $t = 1, 2, \ldots, \|T\|$
k	Index for materials and labor for making a product and $k = 1, 2, \ldots, \|K\|$
r	Index for types of renewable energy sources (e.g., $r = 1$ is for wind, $r = 2$ solar) and $r = 1, 2, \ldots, \|R\|$

Table 2 Model Parameters

Symbol	Description
c_{itm}	Material/labor cost of making product i in period t at plant m
h_{itm}	Unit holding cost for product i in period t at plant m
π_{it}	Backorder cost for product i in period t
β_{im}	Transportation cost for shipping per-unit product i from plant m to the central warehouse
D_{it}	Demand for product i in period t
a_{ikm}	Amount of resource k (other than energy) needed for product i at plant m
b_{ktm}	Available resource k (other than energy) in period t at plant m
$g_{0,rtm}$	Energy price of onsite renewables type r in period t at plant m
$g_{1,rtm}$	Energy price of grid renewables type r in period t at plant m
$g_{2,tm}$	Energy price of conventional energy in period t at plant m
$G_{0,rtm}$	Available onsite renewable energy r in period t at plant m
$G_{1,rtm}$	Available grid renewable energy r in period t at plant m
$G_{2,tm}$	Available conventional energy in period t at plant m
E_{itm}	Energy consumed for making one unit of product i in period t at plant m

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Table 3 Decision Variables and Dependent Variables

Symbol	Description
x_{itm}	Production quantity of product i in period t at plant m
q_{itm}	Inventory level of product i in period t at plant m
v_{itm}	Backorders of product i in period t at plant m
$y_{0,rtm}$	Onsite renewable energy r consumed in period t at plant m
$y_{1,rtm}$	Grid renewable energy r consumed in period t at plant m
$y_{2,tm}$	Conventional energy consumed in period t at plant m

The 1st subscript "0" means onsite renewable energy, "1" means external renewable energy, and "2" means external conventional energy.

to meet the needs of diverse customers. Together, they produce I types of products to meet the aggregate demand over a planning horizon of T periods. Such a multi-plant, multi-product, production planning model, denoted as **P1**, can be formulated as

Problem P1:

$$\min \sum_{m\in M}\sum_{i\in I}\sum_{t\in T}(c_{itm}x_{itm}+h_{itm}q_{itm}+\pi_{it}v_{itm}+\beta_{im}x_{itm}) + \sum_{m\in M}\sum_{t\in T}\left[\sum_{r\in R}(g_{0,rtm}y_{0,rtm}+g_{1,rtm}y_{1,rtm})+g_{2,tm}y_{2,tm}\right] \tag{1a}$$

$$\text{s.t.} \sum_{m\in M}(x_{itm}-q_{itm}+v_{itm})\geq D_{it}, \forall i\in I, \text{ and } t\in\{1\} \tag{1b}$$

$$\sum_{m\in M}(x_{itm}+q_{i,t-1,m}-q_{itm}-v_{i,t-1,m}+v_{itm})\geq D_{it}, \quad \forall i\in I, \text{ and } t\in T\setminus\{1\} \tag{1c}$$

$$\sum_{i\in I}a_{ikm}x_{itm}\leq b_{ktm}, \forall k\in K, t\in T, \text{ and } m\in M \tag{1d}$$

$$\sum_{i\in I}E_{itm}x_{itm}=\sum_{r\in R}(y_{0,rtm}+y_{1,rtm})+y_{2,tm}, \forall t\in T \text{ and } m\in M \tag{1e}$$

$$y_{0,rtm}\leq G_{0,rtm}\ \forall r\in R, m\in M, \text{ and } t\in T \tag{1f}$$

$$y_{1,rtm}\leq G_{1,rtm}\ \forall r\in R, m\in M, \text{ and } t\in T \tag{1g}$$

$$y_{2,tm}\leq G_{2,tm}\ \forall t\in T, \text{ and } m\in M \tag{1h}$$

$$x_{itm},q_{itm},v_{itm},y_{0,rtm},y_{1,rtm},y_{2,tm}\geq 0\ \forall i\in I, r\in R, m\in M, \text{ and } t\in T \tag{1i}$$

The objective (1a) is to minimize the total cost comprised of non-energy and energy-related expenses. Non-energy cost items include production, inventory holding, backorder and transportation costs. Energy costs include generation cost of onsite renewables, and the purchase of grid renewable and conventional energy. Constraints (1b)–(1d) are the fundamental production-inventory balance equations capturing the relation among production, demand, inventory, and backorders. Constraint (1e) is the energy balance equation ensuring that the total energy consumed in period t is equal to the sum of onsite and grid energy, for each plant. Constraints (1f)–(1h) define the supply limitations for both renewable and conventional energy for each plant. Finally, constraint (1i) simply states the nonnegativity requirements for all the decision variables.

If the onsite renewable generation exceeds the total electricity demand in period t, the surplus energy is fed into the utility grid through the net metering mechanism. If this additional revenue to the plant is considered, the term $\Sigma_t\Sigma_r s_{rtm}(G_{0,rtm}-y_{0,rtm})$, where s_{rtm} is the feed-in tariff of surplus renewable energy in plant m, should be subtracted from the objective function (1a). Additionally, the company's investment in different types of renewable generation is reflected by the generation capacity $G_{0,rtm}$.

For plant m, problem **P1** shows the total energy consumed in time period t is equal to $E_{tm}=\Sigma_r(y_{0,rtm}+y_{1,rtm})+y_{2,tm}$ and the GEC criterion for each plant m is measured as $\Sigma_{r\in R}(y_{0,rtm}+y_{1,rtm})/\Sigma_{m\in M}E_{tm}$. If a mandatory GEC level is enforced for the whole manufacturing system such that green energy must account for $100\times\rho\%$ (for $0\leq\rho\leq 1$) of its total energy consumption, the following constraint should be added to problem **P1** for each period

$$\frac{\sum_{r\in R}\sum_{m\in M}(y_{0,rtm}+y_{1,rtm})}{\sum_{m\in M}E_{tm}}\geq\rho,\ \forall t\in T. \tag{2}$$

It is worth mentioning that by modifying constraint (1e), **P1** can accommodate the situations where some plants may share similar onsite generation system due to geographical proximity, or when their electric energy is provided by same utility company.

4. Multistage Stochastic Production Model

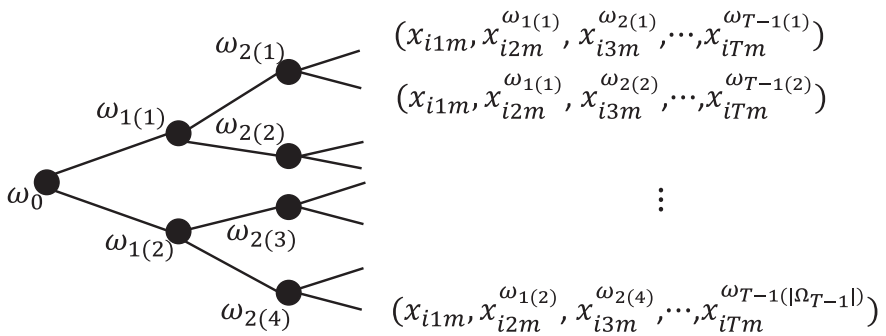
In this section, we explicitly take into account intermittency of wind and solar power by developing a scenario decision tree, and further formulate and solve a multistage stochastic production-inventory model to meet the aggregate demand in each period.

4.1. Scenario Tree for Renewable Energy Generation

We model intermittency of onsite and grid wind and solar generation in each plant m based on a scenario tree. At the beginning of each period, a distributional knowledge of available renewable energy through the scenario tree model is assumed. This modeling approach is adopted based on the fact that the production horizon is long enough so the exact

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Figure 2 Scenario Tree Model for Renewable Energy Generation



throughput of wind and solar generation is not known in the current period. Rather, we assume renewable energy availability for each time period to be captured by scenarios. Let Ω_t denote the set of scenarios for time period t, and each scenario in Ω_t be indexed by ω_t. Each scenario ξ^{ω_t} is associated with probability p^{ω_t}. For consistency of the notation, we also define $\Omega_0 = \{\xi^{\omega_0}\}$ and $p^{\omega_0} = 1$ for $t = 0$. To illustrate the idea, Figure 2 depicts a scenario tree with two scenarios in period 1, and four scenarios in period 2. Note that ω_t in the scenario tree has a corresponding ancestor scenario ω_{t-1}, denoted by $a(\omega_t)$. Additionally, the value of p^{ω_t} is the joint probability of all ancestor scenarios of ω_t on the path from the root ω_0 to ω_t.

Correspondingly, under scenario ω_t, let $G_{0,rm}^{\omega_t}$ and $G_{1,rm}^{\omega_t}$ denote the available onsite and grid renewable energy for type r in period t for plant m, respectively. Clearly, the amount of the available renewable energy is bounded by the generation capacity. Table 4 lists the stochastic parameters associated with the scenario tree.

The decisions associated with production-inventory control need to be made at the beginning of each period. Once the production quantities at a period are decided, the total energy required for production is also determined and the manufacturer can wait until the uncertainties unfold in order to make renewable energy consumption level decisions. The decisions corresponding to each branch from the root to each scenario in the final periods are listed on the right side of Figure 2. The detailed explanation of these decision variables are provided in Table 5.

Table 4 Stochastic Parameters

Symbol	Description
ω_t	A scenario for random renewable energy generation in period t
Ω_t	Set of all possible scenarios ω_t
$a(\omega_t)$	The corresponding ancestor scenario ω_{t-1} of scenario ω_t
$G_{0,rtm}^{\omega_t}$	Available onsite renewable energy r in period t under scenario ω_t in plant m
$G_{1,rtm}^{\omega_t}$	Available external renewable energy r in period t under scenario ω_t in plant m
p^{ω_t}	Probability of scenario ω_t in period t

Table 5 Decision Variables in Scenario Tree

Symbol	Description
$x_{itm}^{\omega_{t-1}}$	Production quantity of product i in period t under scenario ω_{t-1} in plant m
$q_{itm}^{\omega_{t-1}}$	Inventory level of product i at the end of period t under scenario ω_{t-1} in plant m
$v_{itm}^{\omega_{t-1}}$	Backorders of product i at the end of period t under scenario ω_{t-1} in plant m
$y_{0,rtm}^{\omega_t}$	Onsite renewable energy r consumed in period t under scenario ω_t in plant m
$y_{1,rtm}^{\omega_t}$	Grid renewable energy r consumed in period t under scenario ω_t in plant m
$y_{2,tm}^{\omega_t}$	Conventional energy consumed in period t under scenario ω_t in plant m

4.2. Multi-Plant Stochastic Production Planning

To handle these uncertainties, we propose the following stochastic programming model, denoted as problem **P2**, to minimize the expected total costs comprised of energy and non-energy expenses.

Problem P2:

$$\min \sum_{m\in M}\sum_{t\in T}\sum_{\omega_{t-1}} p^{\omega_{t-1}} \left[\sum_{i\in I}\left(c_{itm}x_{itm}^{\omega_{t-1}} + h_{itm}q_{itm}^{\omega_{t-1}} + \pi_{itm}v_{itm}^{\omega_{t-1}} + \beta_{im}x_{itm}^{\omega_{t-1}}\right)\right] + \sum_{m\in M}\sum_{t\in T}\sum_{\omega_t} p^{\omega_t}\left[\sum_{r\in R}\left(g_{0,rtm}y_{0,rtm}^{\omega_t} + g_{1,rtm}y_{1,rtm}^{\omega_t}\right) + g_{2,tm}y_{2,tm}^{\omega_t}\right] \quad (3a)$$

$$\text{s.t.} \sum_{m\in M}\left(x_{itm}^{\omega_0} - q_{itm}^{\omega_0} + v_{itm}^{\omega_0}\right) \geq D_{it}, \forall i \in I, \text{ and } t \in \{1\} \quad (3b)$$

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$$\sum_{m\in M}(x_{itm}^{\omega_{t-1}}+q_{it-1,m}^{a(\omega_{t-1})}-q_{itm}^{\omega_{t-1}}-v_{it-1,m}^{a(\omega_{t-1})}+v_{itm}^{\omega_{t-1}})\geq D_{it}, \quad \forall i\in I, \text{ and } t\in T\setminus\{1\}, \omega_{t-1}\in\Omega_{t-1} \tag{3c}$$

$$\sum_{i\in I}a_{ikm}x_{itm}^{\omega_{t-1}}\leq b_{ktm},\ \forall k\in K,\ \text{and } t\in T, \quad \omega_{t-1}\in\Omega_{t-1} \text{ and } m\in M \tag{3d}$$

$$\sum_{i\in I}E_{itm}x_{itm}^{a(\omega_t)}=\sum_{r\in R}(y_{0,rtm}^{\omega_t}+y_{1,rtm}^{\omega_t})+y_{2,tm}^{\omega_t},\ \forall t\in T, \quad \omega_t\in\Omega_t \text{ and } m\in M \tag{3e}$$

$$y_{0,rtm}^{\omega_t}\leq G_{0,rtm}^{\omega_t},\ \forall r\in R, t\in T, \omega_t\in\Omega_t \text{ and } m\in M \tag{3f}$$

$$y_{1,rtm}^{\omega_t}\leq G_{1,rtm}^{\omega_t},\ \forall r\in R, t\in T, \omega_t\in\Omega_t \text{ and } m\in M \tag{3g}$$

$$y_{2,tm}^{\omega_t}\leq G_{2,tm}\ \forall t\in T, \omega_t\in\Omega_t \text{ and } m\in M \tag{3h}$$

$$x_{itm}^{\omega_{t-1}}, q_{it}^{\omega_{t-1}}, v_{it}^{\omega_{t-1}}\geq 0, \forall i\in I, t\in T, \omega_{t-1}\in\Omega_{t-1} \text{ and } m\in M \tag{3i}$$

$$y_{0,rtm}^{\omega_t}, y_{1,rtm}^{\omega_t}, y_{2,tm}^{\omega_t}\geq 0\ \ \forall r\in R, t\in T, \omega_t\in\Omega_t \text{ and } m\in M \tag{3j}$$

First, at time period $t = 1$, the first-stage decisions on production quality x_{i1m}, inventory level q_{i1m}, backorders v_{i1m} for all $i \in I$ and $m \in M$ are made prior to the realization of any renewable energy scenarios, by constraints (3b), (3d), (3i) (to be consistent, $x_{i1}^{\omega_0}$, $q_{i1}^{\omega_0}$, $v_{i1}^{\omega_0}$ denote x_{i1m}, q_{i1m}, v_{i1m}). Second, depending on the renewable energy scenario realization at the end of $t = 1$ (e.g., $G_{0,r1m}^{\omega_1}$, $G_{1,r1m}^{\omega_1}$), the recourse decisions on $y_{0,r1m}^{\omega_1}$ and $y_{1,r1m}^{\omega_1}$ are limited to the available renewable energy by constraints (3f), (3g), and (3j), and the conventional energy $y_{2,1}^{\omega_1}$ is allocated to meet the energy needs by constraints (3h) and (3j). Constraint (3e) defines the relationship between the first-stage decisions x_{i1m}, q_{i1m}, v_{i1m} and the recourse decisions $y_{0,r1m}^{\omega_1}$, $y_{1,r1m}^{\omega_1}$, $y_{2,1m}^{\omega_1}$ for any scenario $\omega_1 \in \Omega_1$ and $m \in M$. The objective here is to minimize the total costs of production, inventory, backorder and transportation as well as the energy. The latter comprises the generation cost of onsite renewable energy, purchasing of grid renewable and conventional energy under all scenarios $\omega_1 \in \Omega_1$ for plant m, expressed as

$$\sum_i(c_{i1}x_{i1}^{\omega_0}+h_{i1}q_{i1}^{\omega_0}+\pi_{i1}v_{i1}^{\omega_0}) + \sum_{\omega_1}p^{\omega_1}\left[\sum_{r\in R}\left(g_{0,r1}y_{0,r1}^{\omega_1}+g_{1,r1}y_{1,r1}^{\omega_1}\right)+g_{2,1}y_{2,1}^{\omega_1}\right]. \tag{4}$$

Depending on the scenario $\omega_1 \in \Omega_1$, the production-inventory decisions for period $t = 2$ are $x_{i2m}^{\omega_1}$, $q_{i2m}^{\omega_1}$, $v_{i2m}^{\omega_1}$ for all $i \in I$ and $m \in M$, by constraints (3c), (3d) and (3i). The corresponding recourse decisions for this period $y_{0,r2m}^{\omega_2}$, $y_{1,r2m}^{\omega_2}$, $y_{2,2m}^{\omega_2}$ for all $r \in R$ and $m \in M$ are governed by constraints (3e)–(3h) and (3j). By repeating these steps, the actual decisions for each period can be obtained and shown in Table 5.

Additionally, we let $a(\omega_t)$ denote the corresponding ancestor scenario ω_{t-1} of scenario ω_t, and they are used in constraints (3c) and (3e). For constraints (3c) with

$$\sum_{m\in M}\left[x_{itm}^{\omega_{t-1}}+q_{it-1m}^{a(\omega_{t-1})}-q_{itm}^{\omega_{t-1}}-v_{it-1m}^{a(\omega_{t-1})}+v_{itm}^{\omega_{t-1}}\right]\geq D_{it},$$

for any i, t, ω_{t-1}, the production quality $x_{itm}^{\omega_{t-1}}$, inventory level $q_{itm}^{\omega_{t-1}}$, and backorders $v_{itm}^{\omega_{t-1}}$ in period t under scenario ω_{t-1}, are also related to their counterparts in period $t-1$ under the scenario where ω_{t-1} is derived, that is, the ancestor $a(\omega_{t-1})$ of ω_{t-1}. For constraints (3e), the total energy consumption under $\omega_t \in \Omega_t$, expressed by $\Sigma_{r\in R}(y_{0,rtm}^{\omega_t}+y_{1,rtm}^{\omega_t})+y_{2,tm}^{\omega_t}$ are imposed for recourse decision on this period's production decision $x_{it}^{a(\omega_t)}$ for all plant $m \in M$ and under the ancestor scenario $a(\omega_t)$.

For the production planners, at the beginning of each period t, the manufacturing schedule is operated based on $x_{itm}^{\omega_{t-1}}$, $q_{itm}^{\omega_{t-1}}$, $v_{itm}^{\omega_{t-1}}$ under the realized ω_{t-1}. As resource decisions, the energy consumptions $y_{0,rtm}^{\omega_t}$, $y_{1,rtm}^{\omega_t}$, $y_{2,tm}^{\omega_t}$ in period t are determined and adjusted through one particular realization of $G_{0,rtm}^{\omega_t}$, $G_{1,rtm}^{\omega_t}$, to meet the renewable and conventional energy needs.

Similarly, if a mandatory GEC level is enforced for the whole manufacturing system such that green energy must account for $100 \times \rho\%$ of its total energy consumption, the following constraint should be added to Problem **P2** for each period

$$\frac{\sum_{r\in R}\sum_{m\in M}(y_{0,rtm}^{\omega_t}+y_{1,rtm}^{\omega_t})}{\sum_{m\in M}E_{tm}^{\omega_t}}\geq\rho, \forall t\in T, \omega_t\in\Omega_t,$$

where the total energy consumption in time period t under renewable energy scenario ω_t is $E_{tm}^{\omega_t} = \Sigma_{r\in R}(y_{0,rtm}^{\omega_t}+y_{1,rtm}^{\omega_t})+y_{2,tm}^{\omega_t}$.

In problem **P2**, first-stage decisions, x_{i1m}, q_{i1m}, v_{i1m}, are independent of the realizations of renewable energy and actually implementable. As the time evolves and the actual observation of renewable energy is realized, optimal renewables are determined to satisfy the energy requirement of the first-stage production and the model is updated over the remaining planning horizon and re-optimized on the fly.

Incorporating uncertainty in the problem comes with a huge computational cost arising from multistage stochastic programming. Before presenting the algorithm, we first calculate the value of multistage stochastic programming (VMS) to see the benefits of **P2**. To that end, we adopt the definitions of VMS of

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Huang and Ahmed (2009) for our problem. In the following, we define a two-stage stochastic production-planning model for a multi-plant case:

$$\min \sum_{m\in M}\sum_{t\in T}\sum_{i\in I}(c_{itm}x_{itm}+h_{itm}q_{itm}+\pi_{itm}v_{itm}+\beta_{im}x_{itm}) + \sum_{m\in M}\sum_{t\in T}\sum_{\omega_t} p^{\omega_t}\left[\sum_{r\in R}\left(g_{0,rtm}y^{\omega_t}_{0,rtm}+g_{1,rtm}y^{\omega_t}_{1,rtm}\right) + g_{2,tm}y^{\omega_t}_{2,tm}\right] \tag{5a}$$

$$\text{s.t.} \sum_{m\in M}(x_{itm}-q_{itm}+v_{itm})\geq D_{it},\forall i\in I, \text{ and } t\in\{1\} \tag{5b}$$

$$\sum_{m\in M}(x_{it}+q_{it-1}-q_{it}-v_{it-1}+v_{it})\geq D_{it},\forall i\in I, \text{ and } t\in T\setminus\{1\} \tag{5c}$$

$$\sum_{i\in I} a_{ikm}x_{it}\leq b_{ktm},\forall k\in K, t\in T, \text{ and } m\in M \tag{5d}$$

$$\sum_{i\in I}E_{itm}x_{itm}=\sum_{r\in R}(y^{\omega_t}_{0,rtm}+y^{\omega_t}_{1,rtm})+y^{\omega_t}_{2,tm},\forall t\in T,\omega_t\in\Omega_t, \text{ and } m\in M \tag{5e}$$

$$\text{Constraints } (3f),(3g),(3h)(3j) \tag{5f}$$

$$x_{itm},q_{itm},v_{itm}\geq 0,\forall i\in I, t\in T, \text{ and } m\in M \tag{5g}$$

In Equation (5), the production quantity does not depend on the realizations of the scenarios and hence the model does not have the flexibility of the multistage model. Considering the block separable structure of the multistage model, the two-stage model essentially assumes the whole uncertainty unfolds at once and all energy planning decisions will be made then. Let z^{MS} and z^{TS} denote the optimal value of Problems **P2** and Equation (5), respectively. It is easy to observe that every solution to Equation (5) is also feasible to **P2** with equal objective function value and therefore

$$z^{TS}\geq z^{MS}. \tag{6}$$

Then the VMS is defined as

$$VMS=z^{TS}-z^{MS}, \tag{7}$$

and the relative value of multistage stochastic programming (RVMS) is defined as

$$RVMS=\frac{z^{TS}-z^{MS}}{z^{TS}}. \tag{8}$$

In the numerical experiments of section 6, we will compute VMS and RVMS, and further justify the advantages of the proposed multistage model in **P2**.

5. Multistage Stochastic Programming Algorithms

In this section we propose a decomposition method to solve Problem **P2**. First we show **P2** has so-called block separable structure. Hence it can be transformed into a two-stage stochastic linear program that can be solved by the modified Benders decomposition method. Below we derive the equivalent two-stage stochastic program and provide the algorithmic statement.

5.1. Block Separable Stochastic Programs

Consider T-stage stochastic linear program of the form:

$$\begin{aligned}&\min_{x_1} c_1x_1+E_{\tilde{\xi}_2}[h_2(x_1,\tilde{\xi}_2)]\\ &\text{s.t.} A_1x_1\leq b_1\\ &\quad x_1\geq 0\end{aligned}$$

where for $t = 2,3,\ldots,T$,

$$\begin{aligned}&h_t(x_{t-1},\tilde{\xi}_t)=\min_{x_t}\tilde{c}_tx_t+E_{\tilde{\xi}_{t+1}}[h_{t+1}(x_t,\tilde{\xi}_{t+1})]\\ &\text{s.t. } A_tx_t\leq\tilde{B}_tx_{t-1}+\tilde{b}_t\\ &x_t\geq 0,\end{aligned}$$

where x_t, for $t = 1, 2, \ldots, T$ are decision variables in stage t, c_1, b_1, and A_t for $t = 1, 2, \ldots, T$ are deterministic parameters, and $\tilde{c}_t$, $\tilde{b}_t$, $\tilde{B}_t$ for $t = 2, 3, \ldots, T$ are possibly stochastic parameters. Assuming a finite scenario tree, the random parameters are defined over a finite probability space $(\Omega, \sigma(\Omega), \mathbb{P})$, where $\Omega = \Omega_1 \times \Omega_2 \times \cdots \times \Omega_T$, where Ω_t is the support of random vector of parameters $\xi_t^{\omega_t}$ at stage t.

If one views a multistage stochastic program as a set of subproblems at each node of a decision tree corresponding to the scenario tree, then block separability means that the decision vector in any subproblem is formed with two sets of separable variables, with separable corresponding objective function and constraints in the mathematical sense. Therefore, the equivalent two-stage stochastic program posseses a first-stage with the extensive form of the aggregate level problems. The value function of the second-stage subproblem corresponding to a realization is the sum of detailed level recourse functions for that realization of the random vector and all its successors. In the following, we formally define a block separable MSLP due to Louveaux (1986).

Definition 1. *A multistage stochastic program has block separable recourse if for all stages $t = 1, \ldots, T$, the decision vectors x_t can be separated into two subvectors*

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$x_t = [x_{1,t}, x_{2,t}]$, *where* $x_{1,t}$ *represents aggregate level decisions and* $x_{2,t}$ *represents detailed level decision, in such a way that*

(i) The objective function is separable in two parts: $c_t x_t = c_{1,t}x_{1,t} + c_{2,t}x_{2,t}$.

(ii) The constraint matrix A_t *is block diagonal:* $A_t = \begin{bmatrix} A_{1,t} & 0 \\ 0 & A_{2,t} \end{bmatrix}$, *where* $A_{1,t}$ *is associated with vector* $x_{1,t}$ *and* $A_{2,t}$ *with vector* $x_{2,t}$.

(iii) The fixed vector b_1, *and random components of constraints,* $\tilde{\mathbf{B}}_\mathbf{t}$ *and* $\tilde{\mathbf{b}}_\mathbf{t}$ *are decomposed to conform with the* $[x_{1,t}, x_{2,t}]$ *separation such that* $\tilde{\mathbf{B}}_\mathbf{t}$ *only possesses nonzero elements in the columns associated with the aggregate level decisions:* $\tilde{\mathbf{B}}_\mathbf{t} = \begin{bmatrix} \tilde{\mathbf{B}}_{\mathbf{1,t}} & 0 \\ \tilde{\mathbf{B}}_{\mathbf{2,t}} & 0 \end{bmatrix}$ *and* $\tilde{\mathbf{b}}_\mathbf{t} = \begin{bmatrix} \tilde{\mathbf{b}}_{\mathbf{1,t}} \\ \tilde{\mathbf{b}}_{\mathbf{2,t}} \end{bmatrix}$.

LEMMA 1. *The multistage stochastic problem* ***P***2 *has block separable structure.*

PROOF. Conforming to the notation in Definition 1, let the aggregate and detailed level decisions be defined as

$$x_{1,t} = \begin{bmatrix} [x_{itm}]_{|I||M|\times 1} \\ [q_{itm}]_{|I||M|\times 1} \\ [v_{itm}]_{|I||M|\times 1} \end{bmatrix}, \quad x_{2,t} = \begin{bmatrix} [y_{0,rtm}]_{|R||M|\times 1} \\ [y_{1,rtm}]_{|R||M|\times 1} \\ [y_{2,tm}]_{|M|\times 1} \end{bmatrix}, \quad \forall t \in T.$$

For the sake of consistency, we change all the constraints into "≤" form. Then define

$$A_{1,1} = \begin{bmatrix} -D & D & -D \\ F & 0 & 0 \end{bmatrix},$$

where $\begin{bmatrix} D = I \cdots & I_{|I|\times|I||M|} \end{bmatrix}$, and F is the block-diagonal matrix with diagonal blocks

$$F_m = \begin{bmatrix} a_{11m} & a_{21m} & \dots & a_{|I|1m} \\ & \vdots & & \\ a_{1|K|m} & a_{2|K|m} & \dots & a_{|I||K|m} \end{bmatrix}, \text{ for } m = 1, \dots, |M|$$

and

$$\tilde{\mathbf{b}}_{\mathbf{1,t}} = \begin{bmatrix} -[D_{i1}]_{|I|\times 1} \\ [b_{k1m}]|K||M| \times 1 \end{bmatrix}, \forall t \in T,$$

where $\tilde{\mathbf{b}}_{\mathbf{1,t}}$ corresponds to the first-stage production constraints defined in Equations (3b) and (3d).
For every $\omega_t \in \Omega_t$,

$$\mathbf{B}_{\mathbf{1,t}}^{a(\omega_t)} = \begin{bmatrix} 0 & D & -D \\ 0 & 0 & 0 \end{bmatrix}, A_{1,t} = \begin{bmatrix} -D & D & -D \\ F & 0 & 0 \end{bmatrix}, \forall t = 2, 3, \dots, T.$$

Given aggregate level decisions $x_{1,t}^{\omega_{t-1}}$, detailed level decisions, $x_{2,t}^{\omega_t}$ will be made by the second-stage problem. In addition, to conform with "≤" sign of the constraints, we replace the equality constraint (5e) by two "≤" constraint. Consequently, the second part of the tender matrices are defined as

$$\mathbf{B}_{\mathbf{2,t}}^{a(\omega_t)} = \begin{bmatrix} E & 0 & 0 \\ E & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{b}_{\mathbf{2,t}}^{\omega_t} = \begin{bmatrix} 0 \\ 0 \\ G_{0,t}^{\mathcal{M},\omega_t} \\ G_{1,t}^{\mathcal{M},\omega_t} \\ G_{2,t}^{\mathcal{M}} \end{bmatrix}, \text{ and}$$

$$A_{2,t} = \begin{bmatrix} 1_{(1\times(2|M|(|R|+1))} \\ 1_{(1\times(2|M|(|R|+1))} \\ I_{(2|M|(|R|+1))\times(2|M|(|R|+1)} \end{bmatrix}, \quad \forall t \in T$$

where $E = \begin{bmatrix} E_1 & E_2 & \cdots E_{|M|} \end{bmatrix}_{1\times|I||M|}$, $E_m = [E_{1tm} \cdots E_{|I|tm}]_{1\times|I|}$,

$$G_{0,t}^{\mathcal{M},\omega_t} = \begin{bmatrix} G_{0,1t1}^{\omega_t} \\ \vdots \\ G_{0,|R|t|M|}^{\omega_t} \end{bmatrix}_{|R||M|\times 1}, \quad G_{1,t}^{\mathcal{M},\omega_t} = \begin{bmatrix} G_{1,1t1}^{\omega_t} \\ \vdots \\ G_{1,|R|t|M|}^{\omega_t} \end{bmatrix}_{|R||M|\times 1},$$

$$G_{2,t}^{\mathcal{M},\omega_t} = \begin{bmatrix} G_{2,t1}^{\omega_t} \\ \vdots \\ G_{t|M|}^{\omega_t} \end{bmatrix}_{|M|\times 1}.$$

Moreover, the objective function can be decomposed into two parts:

$$c_{1,t} = [c_{1t1}, \dots, c_{|I|t|M|}, h_{1t1}, \dots, h_{|I|t|M|}, \pi_{1t1}, \dots, \pi_{|I|t|M|}, \beta_{11}, \dots, \beta_{|I||M|}]^T$$
$$c_{2,t} = [c_{1t1}, \dots, c_{|I|t|M|}, h_{1t1}, \dots, h_{|I|t|M|}, \pi_{1t1}, \dots, \pi_{|I|t|M|}, \beta_{11}, \dots, \beta_{|I||M|}]^T$$

□

As studied in Louveaux (1986), if the problem possesses the block separable structure, it can be converted into an equivalent two-stage stochastic programming problem. For the problem **P2**, we present its equivalent formulation under the modified Benders decomposition algorithm. The first stage in its equivalent model determines the production quantities, the inventory levels and the backorders in each period, while the second stage decides the energy consumption per category in each period t under scenario ω_t.

5.2. Algorithm Description

In the following, the modified Benders decomposition method for the equivalent two-stage stochastic

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program is presented, and several problems are also formulated, including the master problem of the first stage in the two-stage equivalent formulation, and a recourse subproblem for each period t and each scenario ω_t.

Master Problem (MP):

$$\min \sum_{m\in M}\sum_{t\in T}\sum_{\omega_{t-1}} p^{\omega_{t-1}}\Bigg[\sum_{i\in I}(c_{itm}x_{itm}^{\omega_{t-1}} + h_{itm}q_{itm}^{\omega_{t-1}} + \pi_{itm}v_{itm}^{\omega_{t-1}} + \beta_{im}x_{itm}^{\omega_{t-1}})\Bigg] + \sum_{t\in T}\sum_{\omega_t} p^{\omega_t}\theta_{tm}^{\omega_t} \tag{9a}$$

$$\text{s.t. } \sum_{m\in M}\left(x_{itm}^{\omega_0} - q_{itm}^{\omega_0} + v_{itm}^{\omega_0}\right) \geq D_{it}, \forall i \in I, \text{ and } t \in \{1\} \tag{9b}$$

$$\sum_{m\in M}\left(x_{itm}^{\omega_{t-1}} + q_{it-1,m}^{a(\omega_{t-1})} - q_{itm}^{\omega_{t-1}} - v_{it-1,m}^{a(\omega_{t-1})} + v_{itm}^{\omega_{t-1}}\right) \geq D_{it}, \quad \forall i \in I, \text{ and } t \in T\setminus\{1\}, \omega_{t-1}\in\Omega_{t-1} \tag{9c}$$

$$\sum_{i\in I} a_{ikm}x_{itm}^{\omega_{t-1}} \leq b_{ktm}, \forall k \in K, \text{ and } t \in T, \omega_{t-1}\in\Omega_{t-1}, m \in M \tag{9d}$$

$$\theta_{tm}^{\omega_t} \geq H_{tm}^{\omega_t}(x_{itm}^{a(\omega_t)}), \forall t \in T, \omega_t \in \Omega_t, m \in M \tag{9e}$$

$$x_{itm}^{\omega_{t-1}}, q_{itm}^{\omega_{t-1}}, v_{itm}^{\omega_{t-1}}, \geq 0, \ \forall i \in I, t \in T, \omega_{t-1}\in\Omega_{t-1}, m \in M, \tag{9f}$$

where $\theta_{tm}^{\omega_t} \geq H_{tm}^{\omega_t}(x_{itm}^{a(\omega_t)})$ is corresponding to an optimality cut generated in the recourse subproblem.

Recourse Problem for Stage t, Scenario ω_t, Plant m (RP(t, ω_t, m)):

$$H_{tm}^{\omega_t}(x_{itm}^{a(\omega_t)}) = \min \sum_{r\in R}\left(g_{0,rtm}y_{0,rtm}^{\omega_t} + g_{1,rtm}y_{1,rtm}^{\omega_t}\right) + g_{2,tm}y_{2,tm}^{\omega_t} \tag{10a}$$

$$\text{s.t. } \sum_{r\in R}(y_{0,rtm}^{\omega_t} + y_{1,rtm}^{\omega_t}) + y_{2,tm}^{\omega_t} = \sum_{i\in I} E_{itm}x_{itm}^{a(\omega_{t-1})} \tag{10b}$$

$$y_{0,rtm}^{\omega_t} \leq G_{0,rtm}^{\omega_t}, \forall r \in R \tag{10c}$$

$$y_{1,rtm}^{\omega_t} \leq G_{1,rtm}^{\omega_t,}\forall r \in R \tag{10d}$$

$$y_{2,tm}^{\omega_t} \leq G_{2,tm} \tag{10e}$$

$$y_{0,rtm}^{\omega_t}, y_{1,rtm}^{\omega_t}, y_{2,tm}^{\omega_t} \geq 0, \forall r \in R \tag{10f}$$

Let $\mu_{tm}^{\omega_t}$, $\lambda_{0,rtm}^{\omega_t}$, $\lambda_{1,rtm}^{\omega_t}$, $\lambda_{2,tm}^{\omega_t}$ be the dual variables associated with above constraints in RP(t, ω_t, m), respectively, the dual of the recourse problem is given as follows:

Dual for Recourse Problem for Stage t Scenario ω_t (DRP(t, ω_t, m)):

$$\max\left(\sum_{i\in I} E_{itm}x_{i,tm}^{a(\omega_t)}\right)\mu_{tm}^{\omega_t} + \sum_{r\in R}(G_{0,rtm}^{\omega_t}\lambda_{0,rtm}^{\omega_t} + G_{1,rtm}^{\omega_t}\lambda_{1,rtm}^{\omega_t}) + G_{2,tm}\lambda_{2,tm}^{\omega_t} \tag{11a}$$

$$\text{s.t. } \mu_{tm}^{\omega_t} + \lambda_{0,rtm}^{\omega_t} \leq g_{0,rtm}, \forall r \in R \tag{11b}$$

$$\mu_{tm}^{\omega_t} + \lambda_{1,rtm}^{\omega_t} \leq g_{1,rtm}, \forall r \in R \tag{11c}$$

$$\mu_{tm}^{\omega_t} + \lambda_{2,tm}^{\omega_t} \leq g_{2,tm} \tag{11d}$$

$$\lambda_{0,rtm}^{\omega_t}, \lambda_{1,rtm}^{\omega_t}, \lambda_{2,tm}^{\omega_t} \leq 0, \forall r \in R \tag{11e}$$

In the following, we present the results that we will exploit to solve the second-stage subproblems' duals and primals heuristically, and therefore avoid solving the linear programs to obtain the optimality cuts used in the algorithm.

LEMMA 2. (OPTIMAL SOLUTION FOR SECOND-STAGE SUB-PROBLEMS' PRIMAL RP(t, ω_t, m)). *Suppose* $c_{tm}^{(1)} \leq c_{tm}^{(2)} \leq \cdots \leq c_{tm}^{(2r+1)}$ *denote the corresponding energy costs* $g_{0,rtm}, g_{1,rtm}, g_{2,tm}$ *in an increasing order in time period* t, *and assume that* $y_{tm}^{(j)\omega_t}$ *and* $G_{tm}^{(j)\omega_t}$ *for* $j = 1, 2, \ldots, 2|M|(|R| + 1)$ *denote corresponding recourse decision and energy capacity parameters, respectively. Moreover, suppose the conventional energy supply is large enough so that all required energy can be satisfied. Then, given the relaxed master problem optimal solution* $x_{itm}^{*a(\omega_t)}$ *and defining* $G_{tm}^{(0)} = 0$, *there exists* $p \in \{0, 1, \ldots, 2|M|(|R| + 1)\}$ *such that* $\Sigma_{j=0}^{p}G_{tm}^{(j)} \leq \Sigma_i E_{itm}x_{itm}^{*a(\omega_t)} < \Sigma_{j=0}^{p+1}G_{tm}^{(j)}$ *and the optimal solution for subproblem in stage* t *scenario* ω_t *will be obtained by* $y_{tm}^{*(j)\omega_t} = G_{tm}^{(j)\omega_t}$ *for* $j = 1, 2, \ldots, p$ *and,* $y_{tm}^{*(p+1)\omega_t} = \Sigma_i E_{itm}x_{itm}^{*\omega_{t-1}} - \Sigma_{j=0}^{p}G_{tm}^{(j)}$, *and* $y_{tm}^{*(j)\omega_t} = 0$ *for* $j = p + 2, p + 3, \ldots, 2|M|(|R| + 1)$. *Hence the optimal solutions can be recovered from this.*

PROOF. Suppose the optimal solution to the relaxed master problem is $x_{itm}^{*a(\omega_t)}$. Then energy constraint (10b) needs to be satisfied by the energy variables $y_{0,rtm}^{\omega_t}$, $y_{1,rtm}^{\omega_t}$, and $y_{2,tm}^{\omega_t}$. Therefore by sorting the corresponding objective function coefficients, as long as the available energy upper bounds corresponding to the energy variables will be tight, the value of p will be determined. And therefore $y_{tm}^{*(p+1)\omega_t} = \Sigma_i E_{itm}x_{itm}^{*\omega_{t-1}} - \Sigma_{j=0}^{p}G_{tm}^{(j)}$ will satisfy the remaining energy requirement $\Sigma_i E_{itm}x_{itm}^{*\omega_{t-1}}$ in the first-stage production. □

THEOREM 1. (OPTIMAL SOLUTION FOR SECOND-STAGE SUB-PROBLEMS' DUAL DRP (t, ω_t, m)). *Consider the dual*

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for recourse problem for stage t scenario ω_t (DRP). Corresponding to the ordering in Lemma 2, let $\lambda_{tm}^{(j)\omega_t}$, for $j = 1, 2, \ldots, 2|M|(|R| + 1)$ denote the dual variables $\lambda_{0,rtm}^{\omega_t}$, $\lambda_{1,rtm}^{\omega_t}$, $\lambda_{2,tm}^{\omega_t}$ such that it corresponds to the primal constraint with $G_{tm}^{(i)}$ on the right-hand side. Also let $y_{tm}^{*(j)\omega_t}$ and p be obtained by Lemma 2. Then the optimal dual solution is $\mu_{tm}^{*\omega_t} = c_{tm}^{(p+1)}$, $\lambda_{tm}^{*(j)\omega_t} = c_{tm}^{(j)} - c_{tm}^{(p+1)}$ for $j = 1, 2, \ldots, p$, and $\lambda_{tm}^{*(j)\omega_t} = 0$ for $j = p + 1, p+ 2, \ldots, 2|M|(|R| + 1)$. Therefore, the optimal solutions can be recovered from this.

Proof. We can rewrite the dual problem constraints (11b), (11c), and (11d) with the new indexing as follows,

$$\mu_{tm}^{\omega_t} + \lambda_{tm}^{(j)\omega_t} \le c_{tm}^{(j)} \text{ for } j = 1, 2, \ldots, 2|M|(|R| + 1) \quad (12)$$

Also consider the complementary slackness conditions for the primal-dual pair using the new indexing:

$$\lambda_{tm}^{(j)\omega_t}\left(y_{tm}^{(j)\omega_t} - G_{tm}^{(j)\omega_t}\right) = 0 \text{ for } j = 1, 2, \ldots, 2|M|(|R| + 1) \quad (13a)$$

$$y_{tm}^{(j)\omega_t}\left(\mu_{tm}^{\omega_t} + \lambda_{tm}^{(j)\omega_t} - c_{tm}^{(j)}\right) = 0 \text{ for } j = 1, 2, \ldots, 2|M|(|R| + 1) \quad (13b)$$

Then $\lambda_{tm}^{*(j)} = 0$ for $j = p + 1, p + 2, \ldots, 2|M|(|R| + 1)$ since $y_{tm}^{(j)\omega_t} - G_{tm}^{(j)\omega_t} \neq 0$, and the complementary slackness conditions (13a) are satisfied. Now we need to find solutions satisfying complementary slackness conditions (13b) and feasible to the dual problem. Note that $\mu_{tm}^{\omega_t}$ is unrestricted in sign, $\lambda_{tm}^{(j)\omega_t} \le 0$, and $\mu_{tm}^{\omega_t} \le c_{tm}^{(p+1)}$. Let $\mu_{tm}^{*\omega_t} = c_{tm}^{(p+1)}$ and $\lambda_{tm}^{*(j)\omega_t} = c_{tm}^{(j)} - c_{tm}^{(p+1)} \le 0$ for $j = 1, 2, \ldots, p$, then the dual feasibility and complementary slackness conditions (13b) are satisfied. Hence, by complementary slackness theorem, such primal-dual pair solution is optimal.

We remark that the dual optimal solution obtained as above has a nice interpretation. Consider the objective function of the dual subproblem and the relation $\Sigma_{i=0}^{p} G_{tm}^{(i)} \le \Sigma_i E_{itm} x_{itm}^{*\omega_{t-1}} < \Sigma_{i=0}^{p+1} G_{tm}^{(i)}$. When the dual subproblem constraint (11b) $\mu_{tm}^{\omega_t}$ is present in all of them, increasing $\mu_{tm}^{\omega_t}$ will result in decreasing $\lambda_{tm}^{(i)}$. Since $\Sigma_{i=0}^{p} G_{tm}^{(i)} \le \Sigma_i E_{itm} x_{itm}^{*\omega_{t-1}}$, such an increase is worthwhile. However, $\mu_{tm}^{\omega_t}$ is bounded by $c_{tm}^{(p+1)}$ due to Equation (13a) and dual feasibility. □

Now we present the modified Benders decomposition algorithm pseudocode to solve problem **P2**.

Algorithm Pseudo code:

Step 0. Define tolerance $\varepsilon \ge 0$. Initialize the set of cuts with $\theta_{tm} \ge -\infty$, $\forall t \in T$, and $m \in M$. Let $\bar{z} = +\infty$.

Step 1. (Solve Master Problem) Solve the relaxed master problem. Let $\hat{x}_{itm}^{\omega_{t-1}}$, $\hat{q}_{itm}^{\omega_{t-1}}$, $\hat{v}_{itm}^{\omega_{t-1}}$, $\forall i \in I$, $m \in M$ and $\omega_t \in \Omega_t$, $t = 1, \ldots, T$, be the optimal solution to the master problem.

Step 2. (Solve Recourse Problem) Solve the recourse problem by Lemma 2 and Theorem 1 for stage t scenario $\omega_t \in \Omega_t$, $t = 1, \ldots, T$, and obtain the optimal primal and dual solutions.

Step 3. (Add Optimality Cut) Calculate $H_{tm}^{\omega_t}(x_{itm}^{*a(\omega_t)}), \forall t \in T, m \in M, \omega_t \in \Omega_t$. If

$$\theta_{tm}^{*} \le (1 - \varepsilon) \sum_{t \in T} \sum_{\omega_t} p^{\omega_t} H_{tm}^{\omega_t}(x_{itm}^{*a(\omega_t)}),$$

the following optimality cut should be added,

$$\theta_{tm} \ge \sum_{t \in T} \sum_{\omega_t} p^{\omega_t} \left[\sum_i E_{itm} x_{itm}^{\omega_{t-1}}\right] \mu_{tm}^{*\omega_t} + \sum_{t \in T} \sum_{\omega_t} p^{\omega_t} \left[\sum_r (G_{0,rtm}^{\omega_t} \lambda_{0,rtm}^{*\omega_t} + G_{1,rtm}^{\omega_t} \lambda_{1,rtm}^{*\omega_t}) + G_{2,tm}^{\omega_t} \lambda_{2,tm}^{*\omega_t}\right]$$

Step 4. (Convergence Check) If no optimality cuts are added, then STOP.

Remark. We assume the conventional energy sources are large enough so that the gap between renewables and actual demand can be satisfied. Hence all recourse problems are feasible and no feasibility cut is required for the algorithm. The assumption of sufficient conventional energy is realistic given the fact that the power grid is built with redundant and backup generating units. Moreover, since the objective here is not to simply satisfy energy requirements but achieve a desirable level of green energy penetration, intermittency of wind and solar generation becomes a key challenge to overcome.

6. Numerical Experiments

In this section, we perform a numerical experiment to demonstrate the application and the performance of the proposed production planning models **P1** and **P2** under renewables integration. The LP formulations

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are implemented in C++ using CPLEX 12.6 via IBM's Concert Technology library. All numerical examples were solved on a Linux workstation with 4 Intel(R) Core(TM) 2 CPU 2.40 GHz processors and 32 GB RAM.

6.1. Two-Plant Deterministic Production-Inventory Planning

We first consider the production planning for a manufacturing system comprised of two plants each being integrated with onsite and grid renewable energy, respectively. The related parameters are set as follows: $|I| = 2$ for two types of products, $|T| = 30$ for 30 planning periods, $|K| = 1$ for one type of non-energy resource, and $|R| = 2$ for two types of renewable resourse ($r = 1$ is for wind, $r = 2$ for solar). The production and energy costs are provided in Table 6. The demands for two product types as well as the available resources are presented in Table 7.

Figures 3 and 4 show the production, inventory, and backorder levels for two product types in two plants under the settings in Tables 6 and 7.

Figure 5 shows how the optimal production and energy cost increases with the GEC level. Given the current setting, the chart indicates that the manufacturer needs to pay more energy bills in order to achieve a higher level of GEC performance.

6.2. Single Plant Multistage Stochastic Production Planning

Different from section 6.1, we consider the uncertainties of renewable energy supply. In the deterministic planning, we assume the availability of renewable energy is certain, so essentially we solve a one-scenario linear programming model. In this section, we test the production-inventory model for a single plant with uncertain onsite and grid renewable energy integration. The related parameters are set as follows: $|I| = 2$ for two types of products, $|K| = 1$ for one type of non-energy resource, and $|R| = 2$ for two types of renewable resourse ($r = 1$ is for wind, $r = 2$ for solar). Other related parameters are presented in Tables 8 and 9 "(since $|M|=1$, we omit the index m)". For this single plant case, the transportation cost is not included.

In the following, we investigate the efficiency of our proposed algorithm for the multistage stochastic production planning model for a single plant. We consider six cases for $|T| = 2, 4, 6, 8, 10$ and 12 for different planning horizons. Each stage (or period) corresponds to 12 hours, and therefore a 12-stage model represents a planning horizon of 6 days.

We generate the scenario tree as follows. At an odd stage corresponding to the daytime, both wind and solar energy are available, and each having two realizations independent from each other. At an even stage corresponding to the night, only wind energy is available. We also assumed the onsite and grid renewable energy of same type follow the same

Table 6 Problem Settings for Two Plants

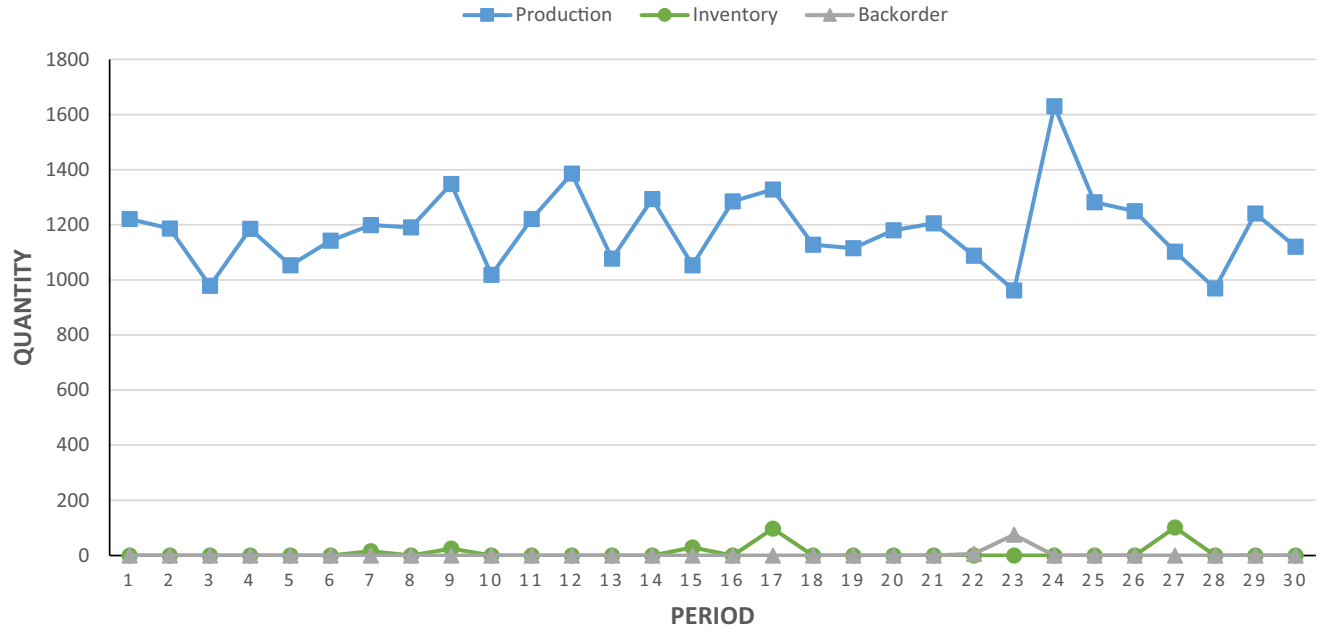
Symbol	Description
c_{itm}	$c_{1t1} = \$80$/unit and $c_{2t1} = \$180$/unit for plant 1, and $c_{1t2} = \$100$/unit and $c_{2t2} = \$200$/unit for plant 2
h_{itm}	$h_{1t1} = h_{1t2} = \$10$/unit/period and $h_{2t1} = h_{2t2} = \$20$/unit/period
π_{itm}	$\pi_{1t1} = \pi_{1t2} = \$1$/unit and $\pi_{2t1} = \pi_{2t2} = \$1$/unit
β_{im}	$\beta_{11} = \$4$/unit, $\beta_{21} = \$9$/unit, $\beta_{12} = \$3$/unit, and $\beta_{22} = \$8$/unit
a_{ikm}	$a_{1k1} = 1$ for product type 1, and $a_{1k1} = 3$ for product type 2 at plant 1, and $a_{1k2} = 1$ for product 1 and $a_{1k2} = 3$ for product 2 at plant 2
$g_{0,rtm}$	$g_{0,1t1} = \$0.05$/kWh, $g_{0,2t1} = \$0.15$/kWh at plant 1, and $g_{0,1t2} = \$0.13$/kWh, $g_{0,2t2} = \$0.07$/kWh at plant 2
$g_{1,rtm}$	$g_{1,1t1} = \$0.06$/kWh, $g_{1,2t1} = \$0.07$/kWh at plant 1, and $g_{1,1t2} = \$0.10$/kWh, $g_{1,2t2} = \$0.09$/kWh at plant 2
$g_{2,tm}$	$g_{2,t1} = \$0.12$/kWh at plant 1, and $g_{2,t2} = \$0.08$/kWh at plant 2
E_{itm}	$E_{1t} = 200$ kWh/unit, and $E_{2t} = 400$ kWh/unit
$G_{0,rtm}$	$G_{0,1t1} = 30,555$ kWh, $G_{0,2t1} = 20,760$ kWh at plant 1, and $G_{0,1t2} = 25,897$ kWh, $G_{0,2t2} = 21,760$kWh at plant 2
$G_{1,rtm}$	$G_{1,1t1} = 35,589$ kWh, $G_{1,2t1} = 55,040$ kWh at plant 1, $G_{1,1t2} = 27,692$ kWh, $G_{1,2t2} = 55,040$ kWh at plant 2 Here we assume that external renewable energy availability is the same across two plants
$G_{2,tm}$	$G_{2,t1} = 395{,}000$ kWh, $G_{2,t2} = 282,000$ kWh

Table 7 Product Demands and Available Resources (each period is 12 hours)

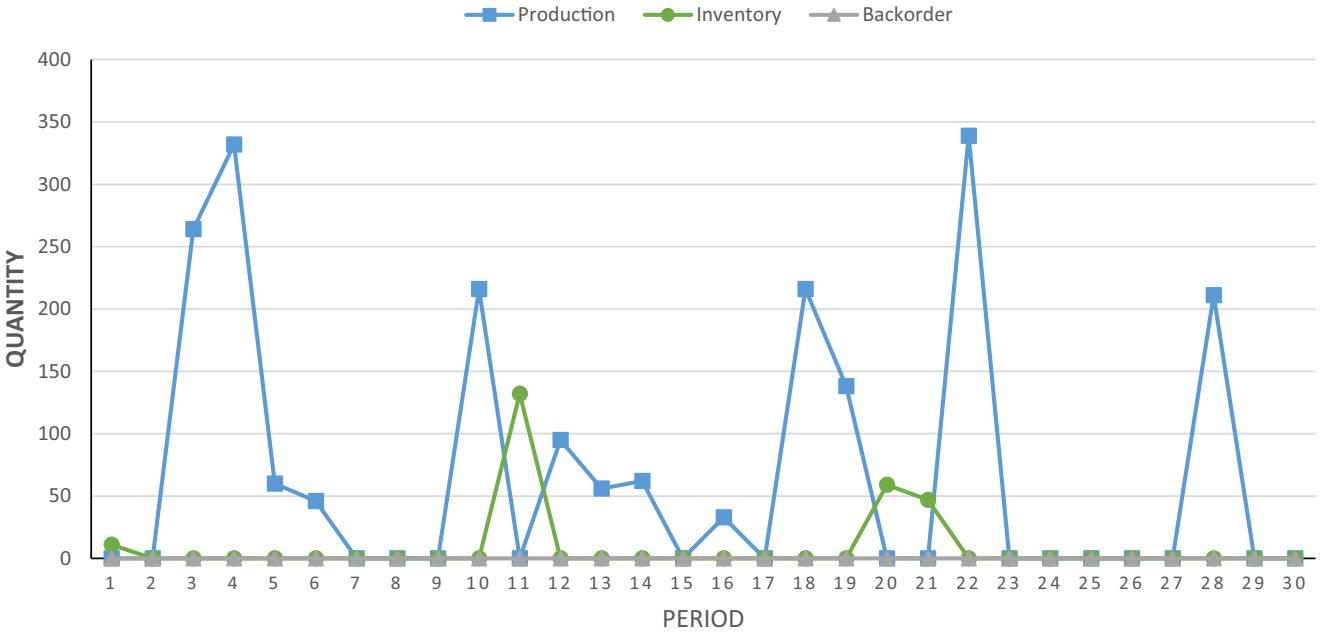
Period t	Aggregate demand for product 1: D_{1t}	Aggregate demand for product 2: D_{2t}	Plant 1 b_{1t1}	Plant 2 b_{1t2}
1	1210	403	1362	1187
2	1197	372	1186	1218
3	1243	269	979	984
4	1517	319	1185	1283
5	1113	327	1053	1186
6	1188	280	1142	1189
7	1184	388	1240	1250
8	1206	350	1191	1079
9	1323	330	1348	1146
10	1260	294	1019	1174
11	1089	389	1221	1180
12	1613	430	1386	1512
13	1133	386	1077	1261
14	1356	429	1294	1349
15	1024	334	1097	958
16	1347	304	1285	1230
17	1231	436	1404	1232
18	1441	300	1128	1245
19	1253	406	1115	1286
20	1121	354	1180	1066
21	1217	426	1205	1219
22	1480	282	1088	1128
23	1031	268	962	857
24	1555	506	1761	1553
25	1282	380	1301	1295
26	1249	406	1286	1377
27	1002	390	1109	1164
28	1281	270	969	1033
29	1241	303	1283	1261
30	1120	403	1269	1351

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Figure 3 Production-Inventory for Product 1 at Two Plants with GEC = 0.35



(a) Product 1 at Plant 1



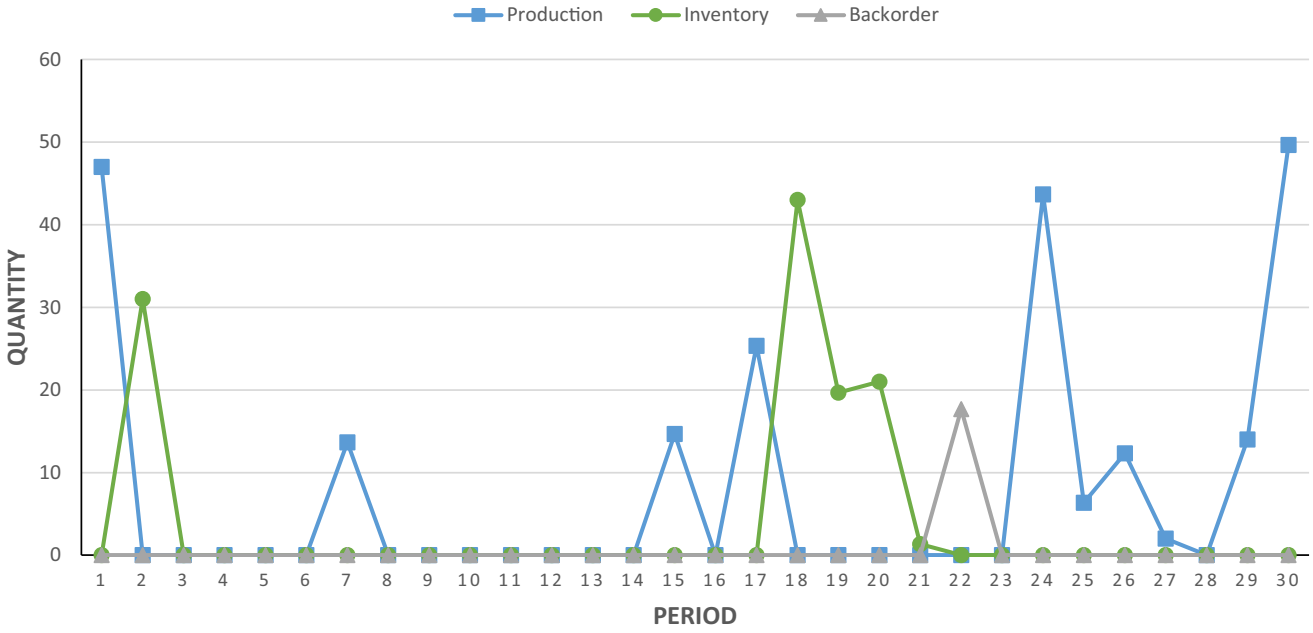
(b) Product 1 at Plant 2

distribution. We compute the joint probabilities at each stage based on the historical data. The sample sizes for different planning periods alongside the computational time for the extensive formulation and the Benders algorithm are reported in Table 10. As the number of stages increases, the numbers of scenarios grow exponentially. Nevertheless, the Benders algorithm shows the efficiency in solving larger problems in a shorter time given limited memory and computational resources.

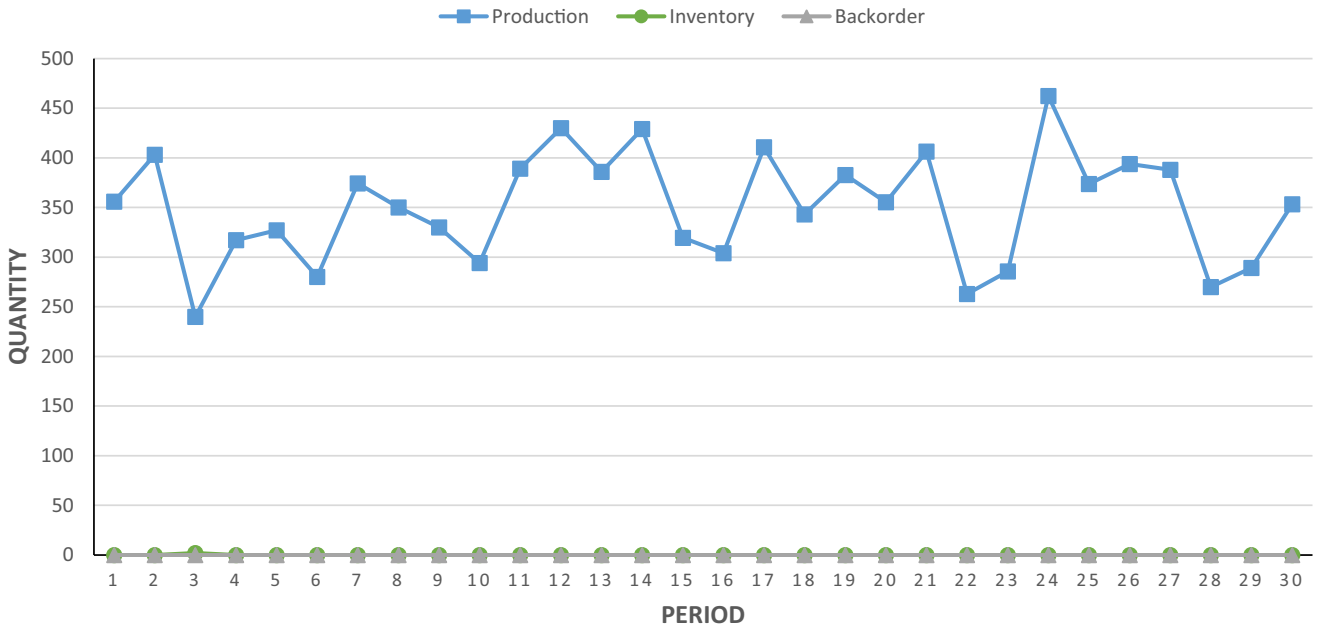
In Table 11, we provide the optimal solutions for two-stage and multistage models with different planning horizons. The numerical results corresponding to RVMS are presented in Figure 6. Our numerical experiments show multistage stochastic program can have a huge impact on minimizing the total

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Figure 4 Production-Inventory for Product 2 at Two Plants with GEC = 0.35



(a) Product 2 at Plant 1



(b) Product 2 at Plant 2

production energy costs if solved for a larger planning horizon assuming the availability of an accurate scenario tree. However, the stochastic nature of renewables generation makes it difficult to have an accurate forecasting of the underlying probability distribution. Hence the decision maker needs to consider the tradeoff between the planning horizon and the potential cost savings.

6.3. Multi-Plant Multistage Stochastic Production Planning

In this section, we investigate the efficiency of our proposed algorithm for the multistage stochastic production planning model for a two-plant case. The related parameters are set as follows: $|I| = 2$ for two product types, $|T| = 2, 3, 4, 5$, and 6 for five cases of planning horizon, $|K| = 1$ for one type of non-energy

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Figure 5 Production Cost with Different Green Energy Coefficient (GEC) Levels

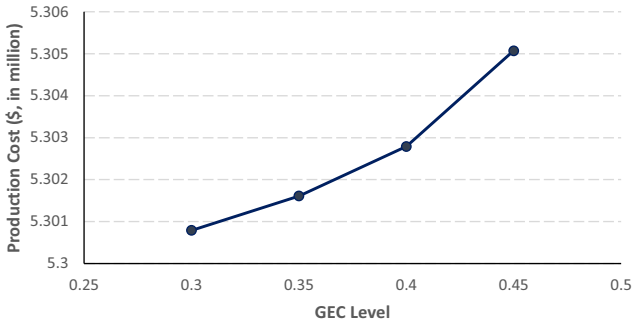


Table 8 Problem Settings for Single Plant

Symbol	Description
c_{it}	c_{1t} = \$10/unit and c_{2t} = \$20/unit
h_{it}	h_{1t} = \$0.5/unit/period and h_{2t} = \$1/unit/period
π_{it}	π_{it} = \$30/unit and π_{2t} = \$50/unit
a_{ik}	a_{1k} = 1 for product type 1, and a_{2k} = 3 for product type 2
$g_{0,rt}$	$g_{0,1}$ = \$0.05/kWh, $g_{0,2t}$ = \$0.1/kWh
$g_{1,rt}$	$g_{1,1}$ = \$0.07/kWh, $g_{1,2t}$ = 0.12/kWh
$g_{2,t}$	$g_{2,t1}$ = \$0.08/kWh
E_{it}	E_{1t} = 2 kWh/unit, and E_{2t} = 4 kWh/unit
$G_{0,rt}$	$G_{0,1t}$ = 58,972 kWh, $G_{0,2t}$ = 39,315 kWh
$G_{1,rt}$	$G_{1,1t}$ = 78,629 kWh, $G_{1,2t}$ = 19,657 kWh
$G_{2,t}$	$G_{2,t}$ = 514,000 kWh

Table 9 Product Demands and Available Resources (each period is 12 hours)

Period t	Aggregate demand for product 1: D_{1t}	Aggregate demand for product 2: D_{2t}	Single plant b_{1t}
1	120,954	40,277	272,466
2	119,722	37,197	237,105
3	124,348	26,886	195,783
4	151,747	31,855	237,066
5	111,339	32,747	210,575
6	118,797	27,962	228,442
7	118,423	38,795	248,045
8	120,565	34,984	238,258
9	132,259	32,995	269,580
10	126,030	29,423	203,874
11	108,873	38,911	244,203
12	161,254	42,988	277,236

Table 10 Number of Scenarios for Different Planning Horizons and Solution Time for Single Plant

Case	Number of stages T	Number of scenarios	Extensive formulation solution time (seconds)	Modified benders algorithm solution time (seconds)
1	2	8	<1	<1
2	4	64	<1	<1
3	6	512	<1	<1
4	8	4096	<1	<1
5	10	32,768	28	3
6	12	262,144	762	120

Table 11 Optimal Values of Two-Stage, Multistage Stochastic Programs and Value of Multistage Stochastic Program (VMS) for Single Plant

Case	Number of stages T	z^{TS}(\$)	z^{MS}(\$)	VMS
1	2	4,317,770	4,317,750	20
2	4	8,539,000	6,860,000	1,679,000
3	6	12,270,000	8,870,000	3,400,000
4	8	16,461,500	10,625,300	3,400,000
5	10	20,609,000	11,010,100	3,400,000
6	12	24,584,000	11,580,000	13,004,000

Figure 6 Relative Value of Multistage Stochastic Programming (RVMS) vs. The Number of Stages for Single Plant

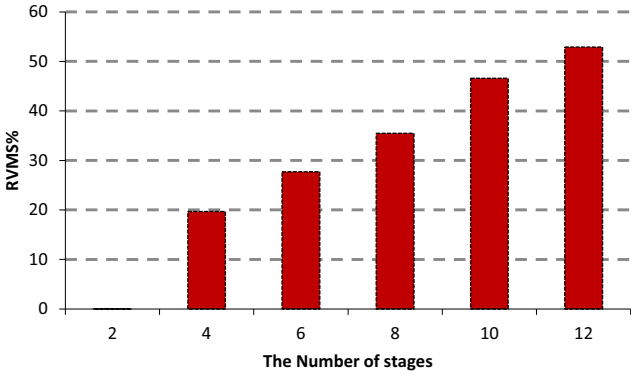


Table 12 Number of Scenarios for Different Planning Horizons and Solution Time for Multiple Plants

Case	Number of stages T	Number of scenarios	Extensive formulation solution time (seconds)	Modified benders algorithm solution time (seconds)
1	2	64	<1	<1
2	3	1024	<1	<1
3	4	4096	<1	<1
4	5	65,526	94	11
5	6	262,144	1031	343

Table 13 Optimal Values of Two-Stage, Multistage Stochastic Programs and Value of Multistage Stochastic Program (VMS) for Multiple Plants

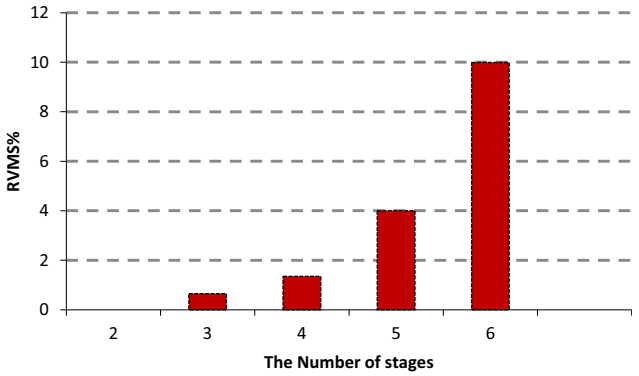
Case	Number of stages T	z^{TS}(\$)	z^{MS}(\$)	VMS
1	2	475,948	475,948	0
2	3	583,421	579,659	3762
3	4	766,507	756,183	10,324
4	5	887,966	852,476	35,490
5	6	1,158,650	1,049,400	109,250

resource, and $|R| = 2$ for two types of renewables recourse. Each stage corresponds to 12 hours, and therefore a 6-stage model represents a planning horizon of 3 days. The deterministic parameters are the same as those in Table 6.

For every plant, the possible realizations of renewables are generated similarly to the single plant case discussed in section 6.2. Additionally, we assume that

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Figure 7 Relative Value of Multistage Stochastic Programming (RVMS) vs. The Number of Stages for Multiple Plants



the realization among different plants are independent, and consequently the scenario tree for the multi-plant corresponds to the joint distribution of renewable energy at each of plants. The sample sizes for different planning periods alongside the computational time for the extensive formulation and the Benders algorithm are reported in Table 12. We provide the optimal solutions for two-stage and multistage models with different planning horizons in Table 13. The numerical results corresponding to RVMS are presented in Figure 7. Again, these results confirm our observations in the single plant case in section 6.2.

7. Conclusions

This study proposes new stochastic programming models and algorithms to optimize a multi-period, multi-plant production-inventory system powered with intermittent wind and solar energy. Under the green quality criterion, the goal is to determine the production quantity, the stock level, and the renewable energy use in each period such that the aggregate production and energy costs are minimized. Though production-inventory systems under demand, capacity and yield uncertainties have been well studied, our model differs appreciably from extant literature in three aspects. First, our study is among the early attempts to penetrate intermittent wind and solar power into the production-inventory system for realizing low-carbon operation. We address the power intermittency issue through a multistage, scenario tree based procedure. Second, we propose to use GEC as a key criterion to assess manufacturing sustainability for which a product is made with a smaller or zero carbon footprint. Third, we develop an efficient modified Benders decomposition algorithm to search for the optimal production and inventory policy using a scenario tree. Our numerical experiments show that achieving a high level of GEC is affordable to the manufacturing community through both onsite and grid renewables integration.

For future work, we are interested in analyzing the sensitivity of the optimal cost and optimal policy with respect to perturbations of scenario tree. Additionally, considering both product demand and the renewable energy uncertainties will require the development of novel models and algorithms.

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References

Apple. 2012. Powering our facilities with clean, renewable energy. Available at http://www.apple.com/environment/renewable-energy (accessed date March 27, 2014).

Armentano, V. A., R. E. Berretta, P. M. Franca. 2001. Lot-sizing in capacitated multi-stage series systems. *Prod. Oper. Manag.* **10**(1): 68–86.

Birge, J. 1982. Decomposition and partitioning methods for multi-stage stochastic linear programs. Technical report 82–6, Department of Industrial and Operations Engineering, University of Michigan.

Birge, J. R., F. Louveaux. 2011. *Introduction to Stochastic Programming*, 2nd edn. Springer, New York, NY.

Chen, C., G. E. Monahan. 2010. Environmental safety stock: The impacts of regulatory and voluntary control policies on production planning, inventory control, and environmental performance. *Eur. J. Oper. Res.* **207**(3): 1280–1292.

Cheng, F., M. Ettl, Y. Lu, D. D. Yao. 2012. A production–inventory model for a push–pull manufacturing system with capacity and service level constraints. *Prod. Oper. Manag.* **21**(4): 668–681.

Duran, S., T. Liu, D. Simchi-Levi, J. L. Swann. 2007. Optimal production and inventory policies of priority and price-differentiated customers. *IIE Trans.* **39**(9): 845–861.

Eberling, B. 2013. Fairfield brewery gets mammoth wind turbine to power plant. Available at http://www.dailyrepublic.com/featured-stories/fairfield-brewery-gets-mammoth-wind-turbine-to-power-plant (accessed date January 19, 2014).

Federgruen, A., P. Zipkin. 1984. Approximation of dynamic, multi-location production and inventory problems. *Management Sci.* **30**(1): 69–84.

Gassmann, H. 1990. Mslip: A computer code for the multistage stochastic linear programming problem. *Math. Program.* **47** (1–3): 407–423.

Geunes, J., L. Retsef, E. Romeijn, D. B. Shmoys. 2011. Approximation algorithms for supply chain planning and logistics problems with market choice. *Math. Program.* **130**(1): 85–106.

Higle, J., K. Kempf. 2011. Production planning under supply and demand uncertainty: A stochastic programming approach. G. Infanger, ed. *Stochastic Programming, International Series in Operations Research & Management Science*, Springer, New York, 297–315.

Huang, K., S. Ahmed. 2009. The value of multistage stochastic programming in capacity planning under uncertainty. *Oper. Res.* **57**(4): 893–904.

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Ierapetritou, M. G., D. Wu, J. Vin, P. Sweeny, M. Chigirinskiy. 2002. Cost minimization in an energy-intensive plant using mathematical programming approaches. *Ind. Eng. Chem. Res.* **41**(21): 5262–5277.

Kall, P., J. Mayer. 2011. *Stochastic Linear Programming*, 2nd edn. Springer, New York, NY.

Karwan, M. H., M. F. Keblis. 2007. Operations planning with real time pricing of a primary input. *Comput. Oper. Res.* **34**(3): 848–867.

Kira, D., M. Kusy, I. Rakita. 1997. A stochastic linear programming approach to hierarchical production planning. *J. Oper. Res. Soc.* **48**(2): 207–211.

Kogan, K. 2009. Production control under uncertainty: Closed-loop versus open-loop approach. *IIE Trans.* **41**(10): 905–915.

Lee, H. L., C. A. Yano. 1988. Production control in multistage systems with variable yield losses. *Oper. Res.* **36**(2): 269–279.

Letmathe, P., N. Balakrishnan. 2005. Environmental considerations on the optimal product mix. *Eur. J. Oper. Res.* **167**(2): 398–412.

Li, H., Y. Tian, F. Chen, T. Jin. 2016. Toward net-zero carbon manufacturing operations: An onsite renewables solution. *J. Oper. Res. Soc.*, DOI: 10.1057/s41274-016-0014-5.

Louveaux, F. V. 1986. Multistage stochastic programs with block-separable recourse. *Math. Program. Study* **28**: 48–62.

Mitra, S., I. E. Grossmann, J. M. Pinto, N. Arora. 2012. Optimal production planning under time-sensitive electricity prices for continuous power-intensive processes. *Comput. Chem. Eng.* **38**(5): 171–184.

Mula, J., R. Poler, J. P. Garca-Sabater, F. C. Lario. 2006. Models for production planning under uncertainty: A review. *Int. J. Prod. Econ.* **103**(1): 271–285.

Rebennack, S. 2014. Generation expansion planning under uncertainty with emissions quotas. *Elec. Power Sys. Res.* **114**: 78–85.

Rebennack, S. 2016. Combining sampling-based and scenario-based nested Benders decomposition methods: Application to stochastic dual dynamic programming. *Math. Program.* **156**(1): 343–389.

Rebennack, S., B. Flach, M. V. F. Pereira, P. M. Pardalos. 2012. Stochastic hydro-thermal scheduling under CO_2 emission constraints. *IEEE Trans. Power Syst.* **27**(1): 58–68.

Sen, S., Z. Zhou. 2014. Multistage stochastic decomposition: A bridge between stochastic programming and approximate dynamic programming. *SIAM J. Optim.* **24**(1): 127–153.

Singer, G., E. Khmelnitsky. 2010. A finite-horizon, stochastic optimal control policy for a production-inventory system with backlog-dependent lost sales. *IIE Trans.* **42**(12): 855–864.

Sinpetru, L. 2014. Honda installs wind turbines at manufacturing plant in Ohio, US. Reported on January 23, 2014. Available at http://news.softpedia.com/news/Honda-Installs-Wind-Turbines-at-Manufacturing-Plant-in-Ohio-US-419798.shtml (accessed date March 28, 2014).

Villarreal, S., J. A. Jimenez, T. Jin, M. Cabrera-Rios. 2013. Designing a sustainable and distributed generation system for semiconductor wafer fabs. *IEEE Trans. Autom. Sci. Eng.* **10**(1): 10–16.

Wang, Y., Y. Gerchak. 1996. Periodic review production models with variable capacity, random yield, and uncertain demand. *Management Sci.* **42**(1): 130–137.

Wee, H., W. Yang, C. Chou, M. V. Padilan. 2012. Renewable energy supply chains, performance, application barriers, and strategies for further development. *Renew. Sustain. Energy Rev.* **16**(8): 5451–5465.

Yano, C. A., H. L. Lee. 1995. Lot sizing with random yields: A review. *Oper. Res.* **43**(2): 311–334.

Zhang, M., S. Küçükyavuz, H. Yaman. 2012. A polyhedral study of multiechelon lot sizing with intermediate demands. *Oper. Res.* **60**(4): 918–935.

Zhang, W., Z. Hua, Y. Xia, B. Huo. 2016. Dynamic multi-technology production-inventory problem with emissions trading. *IIE Trans.* **48**(2): 110–119.

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